Ordinal Deep Learning for Automated Fall Armyworm Damage Severity Scoring in FAWCheck

A CNN-Based Approach to the Prasanna Scale

BS Computer Science | CMSC 190 Proposal Defense | November 2026

2,084
labeled images inherited from FAWCheck v1

4
models compared: softmax, CORAL, CORN, THOR

15
train-and-evaluate runs (5 folds × 3 seeds)

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Fall armyworm threatens Philippine maize

> 5,000

farmers affected, mid-2024

~3,200 ha

infested, mid-2024

Fall Armyworm

Spodoptera frugiperda

734

metric tons of corn destroyed, worth about PHP 57.03 million (mid-August 2024)

−20.35%

second-quarter output; retail corn prices rose 5.75%

*Not only **whether** a plant is infested, but **how badly**.*

Source: Department of Agriculture figures as reported in Casasola, Anacleto & Nocum (2026); manuscript p. 1

Why order matters on the Prasanna scale



Softmax cross-entropy ignores order

3 5 → 6

Predicting 3 or 5 when the true score is 6 costs the **same loss**.

Errors are not interchangeable

5 | 6 7

6 and 7 share the Susceptible band. 5 and 7 cross from **Partially resistant** to **Susceptible**.

Prasanna et al. (2018) scale, bands from Table 1 (p. 10); severe damage defined as scores 6 to 9 (p. 2)

FAWCheck v1 leaves three problems

THIS STUDY
Image CNN

+

Rule-based scoring from a symptom questionnaire

→

Fused severity estimate

1 Order is ignored

76.47%

of CNN predictions land within one level of the true score: the errors are structured, yet softmax treats scores as unrelated.

2 Baseline cannot be trusted

50.00%

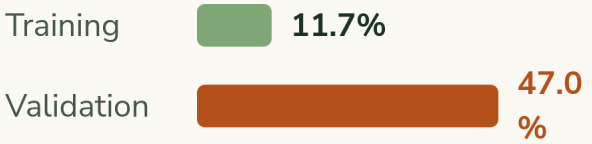
exact accuracy on 68 images (34/68). 95% CI [38.4%, 61.6%], width 23.1 pp.



3 Split is non-representative

47.0%

of validation images are score 1, against 11.7% of training.



Casasola et al. (2026) as reported in manuscript pp. 2 to 3; Table 4 (p. 30); partition audit (p. 28)

Four objectives, one question

GENERAL OBJECTIVE

Does ordinal regression improve automated FAW foliar damage severity scoring over standard multi-class classification, under an evaluation protocol that can support that comparison?

1 **Ordinal vs softmax**
CORAL, CORN and THOR against softmax, on identical folds with identical imbalance handling.

2 **Ordinal vs ordinal**
Do published expectations hold at nine levels under severe class imbalance?

3 **Human reference**
Author and expert agreement on every new label; label consistency across near-duplicates.

4 **Protocol effect**
Rebuild the evaluation protocol and separate its effect from the model's.

Statement of objectives, manuscript pp. 4 to 5

Who benefits from this study

HOW V1 REPORTED PERFORMANCE

50.00%

on **68** images

no interval

FAWCheck users, IPB, Entomology Laboratory Division

A severity model reported with quantified uncertainty: 95% confidence intervals.

Agricultural computer vision

No located work applies rank-consistent ordinal regression to an expert-rated crop damage scale.

Methodology

A reusable evaluation protocol for later FAWCheck iterations and other rating-scale tasks.

Significance of the study, manuscript pp. 5 to 6; gap statement, p. 21

Approach in six steps

1

Audit and rebuild

Rebuild the evaluation protocol

Obj 4

2

Expand and label

More data, plus a Philippine external test set

Obj 3

3

Check labels

Author and expert agreement; near-duplicate consistency

Obj 3

4

Re-measure v1

PyTorch port of the v1 recipe becomes the baseline

Obj 4

5

Compare models

Softmax, CORAL, CORN and THOR

Obj 1 and 2

6

Final test

One evaluation on the external test set

Run once

Every rule is fixed before any ordinal model is trained. Any deviation is reported.

Research design, manuscript pp. 22 to 23

One backbone, four output heads

MobileNetV3Small backbone, shared by all four

Identical folds, imbalance handling, augmentation, tuning budget and computing environment.

Softmax (control)

●●●●●●●●●

9 logits, cross-entropy. Median readout; argmax also reported.

CORAL

●●●●●●●●

8 threshold logits: one shared weight vector, a bias per threshold.

CORN

●●●●●●●●

8 conditional logits, no weight sharing.

THOR

●

1 linear unit, thresholded pairwise ranking loss.

1 Linear probing

Backbone frozen; train the output layer.

2 Fine-tuning

Whole network, lower learning rate.

3 Classifier retraining

Backbone frozen, output layer re-initialized, class-balanced samples.

Table 5 (p. 33); imbalance handling (p. 34); training procedure (p. 35)

Data and evaluation protocol

Data

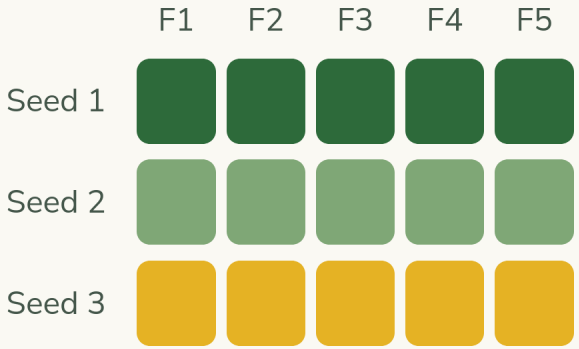
2,084 inherited labeled images

+146 new labels at scores 7 to 9

2,230 planned cross-validation pool

≥ 150 Philippine phone photos: the external test set, evaluated once

Cross-validation



15 runs

5 folds × 3 seeds. Stratified on the merged label, grouped by near-duplicate cluster.

Metrics

Selection: macro-averaged MAE

Quadratic weighted kappa

Severe-damage recall (score 6+)

Band-crossing error rate

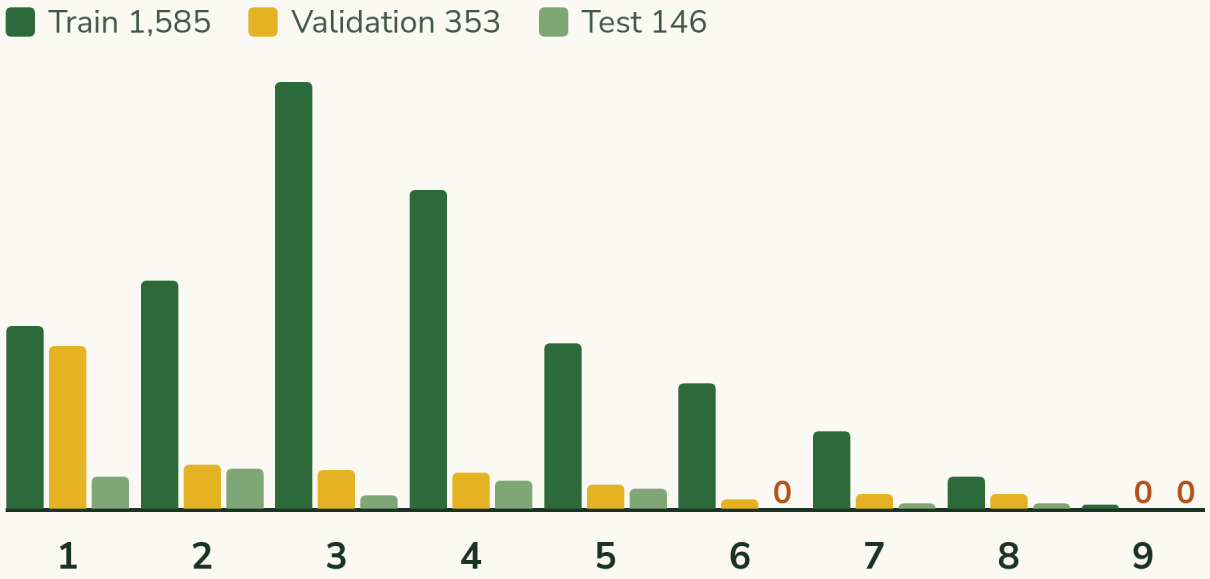
Human reference: author against expert, on the same images.

Score 9 has only **4 images**, so scores 8 and 9 merge into one group of **56** for headline metrics (pre-registered merge rule).

Data sources (pp. 24 to 25); cross-validation (pp. 28 to 29); metrics (pp. 39 to 41); merge rule (p. 35)

The inherited split is discarded, not patched

Images per split, by severity score



1 Not representative

Score 1 is 11.7% of training but 47.0% of validation.

2 Unevaluable classes

Score 6 has no test images. Score 9 never appears in validation or test.

3 Distributions diverge

Scores 5 to 9 are 25.8% of training but 47.1% of v1's 68-image set, about twice as many.

Decision: rebuild with stratified, leakage-controlled folds.

Table B.1 and Appendix B (pp. 54 to 55); why the partition is discarded (p. 28)

Where the study stands and what is next

COMPLETED

- File-level audit of the inherited partition
- 95% CIs recomputed for the v1 results
- Design and pre-registered rules written

NEXT, BEFORE ANY ORDINAL MODEL

- Label 146 more images at scores 7 to 9
- Collect at least 150 phone photos for the external test set
- Cluster near-duplicates and build the folds
- Re-measure v1, run the sensitivity check, fix tuning budget B

AFTER EXPERIMENTS

- Train and compare the four models
- Score final models once on the external test set
- Write Results and Conclusion

Out of scope. Rule-based scoring, fusion, ONNX deployment, cob and ear damage, yield loss.

Limits. Labels describe photographs, not field plants. Author and expert agreement is a reference, not a ceiling.

Scope and limitations, manuscript pp. 6 to 8; sequence from pp. 22 to 23, 38, 42

BACKUP

The Prasanna scale in full



Figure 5. Rating of maize plants based on fol...

Score	Damage symptoms (shortened)	Response
1	No visible leaf-feeding damage	Highly resistant
2	Few pinholes on 1 to 2 older leaves	Resistant
3	Shot-holes on a few leaves (fewer than 5)	Resistant
4	Shot-holes on 6 to 8 leaves; few small lesions (to 1.3 cm)	Partially resistant
5	Lesions over 2.5 cm on 8 to 10 leaves; some irregular holes	Partially resistant
6	Several large lesions and holes on several whorl and furl leaves	Susceptible
7	Many lesions of all sizes on several whorl and furl leaves	Susceptible
8	Many lesions on most whorl and furl leaves; many large holes	Highly susceptible
9	Whorl and furl almost totally destroyed; plant dying	Highly susceptible

Shortened descriptors. Prasanna et al. (2018), Table 3 and Figure 5, modified from Davis and Williams (1992); manuscript Table 1 (p. 10)